

ASSIGNMENT NO. 10: Window Functions and Analytical Queries

Q1) Create a MySQL database named SalesDB and a table named SalesData.

SalesData Table

Column Name	Data Type	Constraint
SaleID	INT	PRIMARY KEY
Salesperson	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	NOT NULL
SaleDate	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL

Insert at least 12 sales records belonging to three different departments.

```
CREATE DATABASE SalesDB;
```

```
USE SalesDB;
```

```
CREATE TABLE SalesData (  
    SaleID INT PRIMARY KEY,  
    Salesperson VARCHAR(50) NOT NULL,  
    Department VARCHAR(30) NOT NULL,  
    SaleDate DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL  
);
```

```
INSERT INTO SalesData  
(SaleID, Salesperson, Department, SaleDate, Amount)  
VALUES  
(1, 'Aarav Shah', 'Electronics', '2026-01-05', 50000.00),  
(2, 'Priya Patil', 'Furniture', '2026-01-08', 40000.00),
```

(3, 'Rohan Desai', 'Electronics', '2026-01-12', 65000.00),
 (4, 'Sneha Joshi', 'Furniture', '2026-01-15', 55000.00),
 (5, 'Karan Mehta', 'Electronics', '2026-02-05', 65000.00),
 (6, 'Meera Kulkarni', 'Furniture', '2026-02-10', 30000.00),
 (7, 'Neha Shah', 'Clothing', '2026-02-12', 45000.00),
 (8, 'Aman Verma', 'Clothing', '2026-02-18', 45000.00),
 (9, 'Divya More', 'Electronics', '2026-03-03', 80000.00),
 (10, 'Rahul Sharma', 'Furniture', '2026-03-06', 60000.00),
 (11, 'Isha Nair', 'Clothing', '2026-03-10', 70000.00),
 (12, 'Om Jadhav', 'Electronics', '2026-03-14', 55000.00);

phpMyAdmin

Server: 127.0.0.1 » Database: salesdb » Table: SalesData

Browse Structure SQL Search Insert Export Import Privileges Operation

Promoting | Edit inline | Edit | Explain SQL | Create PHP code | Refresh

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

			SaleID	Salesperson	Department	SaleDate	Amount
☐	Edit	Copy	Delete	1	Aarav Shah	Electronics	2026-01-05 50000.00
☐	Edit	Copy	Delete	2	Priya Patil	Furniture	2026-01-08 40000.00
☐	Edit	Copy	Delete	3	Rohan Desai	Electronics	2026-01-12 65000.00
☐	Edit	Copy	Delete	4	Sneha Joshi	Furniture	2026-01-15 55000.00
☐	Edit	Copy	Delete	5	Karan Mehta	Electronics	2026-02-05 65000.00
☐	Edit	Copy	Delete	6	Meera Kulkarni	Furniture	2026-02-10 30000.00
☐	Edit	Copy	Delete	7	Neha Shah	Clothing	2026-02-12 45000.00
☐	Edit	Copy	Delete	8	Aman Verma	Clothing	2026-02-18 45000.00
☐	Edit	Copy	Delete	9	Divya More	Electronics	2026-03-03 80000.00
☐	Edit	Copy	Delete	10	Rahul Sharma	Furniture	2026-03-06 60000.00
☐	Edit	Copy	Delete	11	Isha Nair	Clothing	2026-03-10 70000.00
☐	Edit	Copy	Delete	12	Om Jadhav	Electronics	2026-03-14 55000.00

Check all With selected: Edit Copy Delete Export

Q2) Write a MySQL query using ROW_NUMBER() to assign a unique row number to every salesperson.

The result should be arranged by sale amount from highest to lowest.

Required Output

Column
Salesperson
Department
Amount
SalesRowNumber

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

ORDER BY Amount DESC, SaleID ASC

) AS SalesRowNumber

FROM SalesData;

The screenshot shows the phpMyAdmin interface with the following details:

- Server:** 127.0.0.1 » **Database:** salesdb » **Table:** SalesData
- Navigation Panel:** Lists various databases including `collegedb_assignment`, `employeeedb`, `healthcarerecordsdb`, `hospitaldb`, `information_schema`, `juniorcollegedb`, `minicollegedb`, `mysql`, `performance_schema`, `phpmyadmin`, `robotech`, `salesdb` (selected), `schooldb`, `seniorcollegedb`, `smartmartdb`, `supportdb`, `test`, and `vaccinationdb`.
- SQL Query:**

```
SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER ( ORDER BY Amount DESC, SaleID ASC ) AS SalesR
```
- Results:** Showing rows 0 - 11 (12 total, Query took 0.0187 seconds.)
- Table Data:**

Salesperson	Department	Amount	SalesRowNumber
Divya More	Electronics	80000.00	1
Isha Nair	Clothing	70000.00	2
Rohan Desai	Electronics	65000.00	3
Karan Mehta	Electronics	65000.00	4
Rahul Sharma	Furniture	60000.00	5
Sneha Joshi	Furniture	55000.00	6
Om Jadhav	Electronics	55000.00	7
Aarav Shah	Electronics	50000.00	8
Neha Shah	Clothing	45000.00	9
Aman Verma	Clothing	45000.00	10
Priya Patil	Furniture	40000.00	11
Meera Kulkarni	Furniture	30000.00	12

Q3) Write a MySQL query using RANK() to rank salespersons according to their sale amount.

Students having the same sale amount should receive the same rank.

Required Output

Column
Salesperson
Department
Amount
SalesRank

SELECT

Salesperson,

Department,

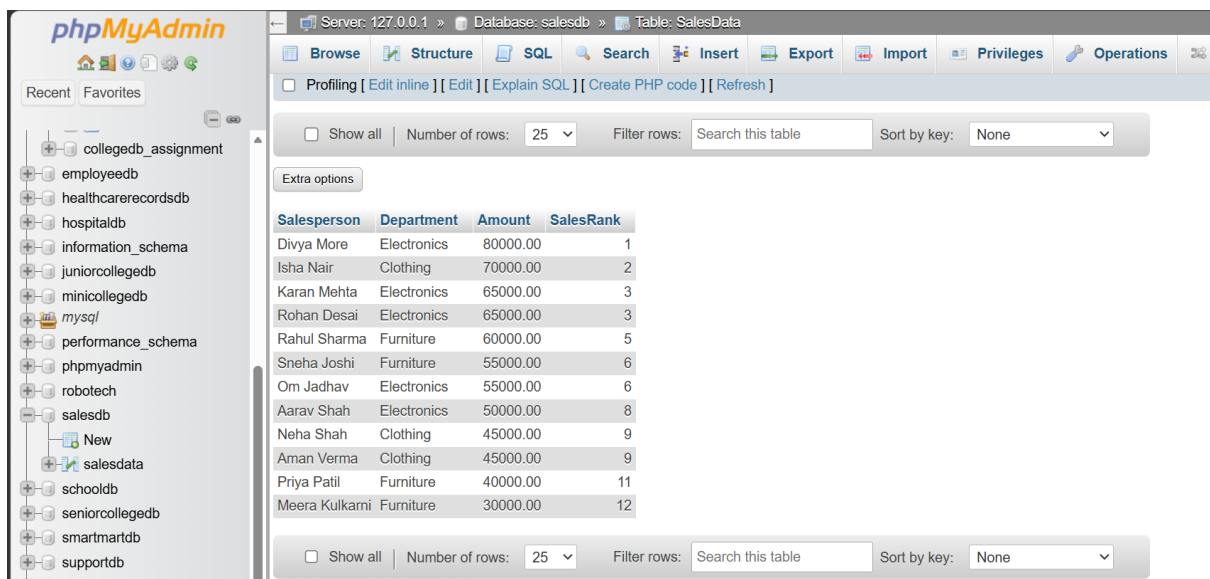
Amount,

RANK() OVER (

ORDER BY Amount DESC

) AS SalesRank

FROM SalesData;



The screenshot shows the phpMyAdmin interface with the 'SalesData' table selected. The table contains 12 rows of data, sorted by 'Amount' in descending order. The columns are Salesperson, Department, Amount, and SalesRank. The SalesRank column shows the rank of each salesperson based on their sale amount, with ties receiving the same rank.

Salesperson	Department	Amount	SalesRank
Divya More	Electronics	80000.00	1
Isha Nair	Clothing	70000.00	2
Karan Mehta	Electronics	65000.00	3
Rohan Desai	Electronics	65000.00	3
Rahul Sharma	Furniture	60000.00	5
Sneha Joshi	Furniture	55000.00	6
Om Jadhav	Electronics	55000.00	6
Aarav Shah	Electronics	50000.00	8
Neha Shah	Clothing	45000.00	9
Aman Verma	Clothing	45000.00	9
Priya Patil	Furniture	40000.00	11
Meera Kulkarni	Furniture	30000.00	12

Q4) Write a MySQL query to rank salespersons separately within each department.

Use:

Required Component
RANK()
OVER
PARTITION BY Department
ORDER BY Amount DESC

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (

PARTITION BY Department

ORDER BY Amount DESC

) AS DepartmentRank

FROM SalesData;

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'salesdb' selected. The main panel displays the 'Table: SalesData' view. A yellow status bar indicates 'Showing rows 0 - 11 (12 total, Query took 0.0013 seconds.)'. Below this, the SQL query is shown: `SELECT Salesperson, Department, Amount, RANK() OVER ( PARTITION BY Department ORDER BY Amount DESC ) AS DepartmentRank FROM SalesData;`. The query results are displayed in a table with 4 columns: Salesperson, Department, Amount, and DepartmentRank. The results are sorted by Department and then by Amount in descending order.

Salesperson	Department	Amount	DepartmentRank
Isha Nair	Clothing	70000.00	1
Neha Shah	Clothing	45000.00	2
Aman Verma	Clothing	45000.00	2
Divya More	Electronics	80000.00	1
Karan Mehta	Electronics	65000.00	2
Rohan Desai	Electronics	65000.00	2
Om Jadhav	Electronics	55000.00	4
Aarav Shah	Electronics	50000.00	5
Rahul Sharma	Furniture	60000.00	1
Sneha Joshi	Furniture	55000.00	2
Priya Patil	Furniture	40000.00	3
Meera Kulkarni	Furniture	30000.00	4

Q5) Write a MySQL query to calculate the running total of sales according to the sale date.

Required Output

Column
SaleID
Salesperson
SaleDate
Amount
RunningTotal

SELECT

SaleID,

Salesperson,

SaleDate,

Amount,

SUM(Amount) OVER (

ORDER BY SaleDate, SaleID

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

) AS RunningTotal

FROM SalesData

ORDER BY SaleDate, SaleID;

phpMyAdmin

Server: 127.0.0.1 » Database: salesdb » Table: SalesData

Browse Structure SQL Search Insert Export Import Privileges Operations

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

		SaleID	Salesperson	SaleDate	Amount	RunningTotal
☐	Edit Copy Delete	1	Aarav Shah	2026-01-05	50000.00	50000.00
☐	Edit Copy Delete	2	Priya Patil	2026-01-08	40000.00	90000.00
☐	Edit Copy Delete	3	Rohan Desai	2026-01-12	65000.00	155000.00
☐	Edit Copy Delete	4	Sneha Joshi	2026-01-15	55000.00	210000.00
☐	Edit Copy Delete	5	Karan Mehta	2026-02-05	65000.00	275000.00
☐	Edit Copy Delete	6	Meera Kulkarni	2026-02-10	30000.00	305000.00
☐	Edit Copy Delete	7	Neha Shah	2026-02-12	45000.00	350000.00
☐	Edit Copy Delete	8	Aman Verma	2026-02-18	45000.00	395000.00
☐	Edit Copy Delete	9	Divya More	2026-03-03	80000.00	475000.00
☐	Edit Copy Delete	10	Rahul Sharma	2026-03-06	60000.00	535000.00
☐	Edit Copy Delete	11	Isha Nair	2026-03-10	70000.00	605000.00
☐	Edit Copy Delete	12	Om Jadhav	2026-03-14	55000.00	660000.00

Check all With selected: Edit Copy Delete Export

Q6) Write a MySQL query to display the highest-performing salesperson from each department.

The query should:

Requirement
Use ROW_NUMBER()
Partition records by department
Arrange sale amounts in descending order
Use a subquery
Display only the first-ranked salesperson from every department

SELECT

Salesperson,

Department,

Amount

FROM (

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

PARTITION BY Department

ORDER BY Amount DESC, SaleID ASC

) AS DepartmentRowNumber

FROM SalesData

) AS DepartmentSales

WHERE DepartmentRowNumber = 1;

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database list including 'salesdb'. The main panel is titled 'Server: 127.0.0.1 » Database: salesdb » Table: SalesData'. It contains a 'Show query box' section with a warning: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this is a green status bar indicating 'Showing rows 0 - 2 (3 total, Query took 0.0017 seconds.)'. The SQL query is displayed in a syntax-highlighted box:

```
SELECT Salesperson, Department, Amount FROM ( SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER ( PARTITION BY BY Amount DESC, SaleID ASC ) AS DepartmentRowNumber FROM SalesData ) AS DepartmentSales WHERE DepartmentRowNumber = 1;
```

 Below the query are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. An 'Extra options' button is also present. The results table has three columns: 'Salesperson', 'Department', and 'Amount'. It contains three rows of data.

Salesperson	Department	Amount
Isha Nair	Clothing	70000.00
Divya More	Electronics	80000.00
Rahul Sharma	Furniture	60000.00